An exponential-time oracle blocks deterministic constructions while permitting quasipolynomial Kt approximation

Shuichi Hirahara, Zhenjian Lu, Hanlin Ren

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Editorial note. Complete author published proof transcribed with original mathematical notation and exact theorem conclusions. The theorem statement prints omega(n), while the surrounding author prose/proof uses omega(1); both are retained as published, without silently editing the author. Display equations may be split across lines for page width; every author term is retained. Original last inequality in approximation proof ends in (1+epsilon)m and is retained.

Allowed standard prerequisites (editorial).

- Hirahara worst-case to average-case pseudorandom generator reconstruction. Original cited generator Lemma56 (cf. reference33) retained with complete parameters; standard earlier construction is not recursively reproved

Definitions and notations (original Sections 1.1, 2.1). For a complexity class $\mathfrak{C}$, a complexity-theoretic statement is $\mathfrak{C}$-relativizing if it is true relative to every oracle $\mathcal{O} \in \mathfrak{C}$.

Definition 20. A language L is in EXPH if there is a constant k, an exponential bound $e(n) = 2^{\mathrm{poly}(n)}$, and a $\mathrm{poly}(e(n))$-time algorithm M such that, for every input $x \in \{0,1\}^*$,

$$x \in L \iff \exists y_1 \forall y_2 \exists y_3 \forall y_4 \dots \exists y_k,\ M(x, y_1, \dots, y_k) = 1,$$

where $y_i \in \{0,1\}^{e(n)}$ for each $i \in [k]$. Equivalently, a language is in EXPH if it can be decided in exponential $(2^{\mathrm{poly}(n)})$ time with access to a PH oracle.

Pseudodeterministic algorithms. A pseudodeterministic algorithm is a probabilistic algorithm that returns a fixed output with high probability on any given input. We will also consider pseudodeterministic oracle (or advice-taking) algorithms. Such an algorithm will maintain its pseudodeterministic behavior (i.e., outputting a fixed answer with high probability for any input) when a correct oracle (or advice) is given.

Dense property (original Section 1.3.2). A property $Q \subseteq \{0,1\}^*$ is dense if for every input length $n \in \mathbb{N}$, we have that $|Q \cap \{0,1\}^n| \geq 2^n / \mathrm{poly}(n)$.

Theorem 23 [68]. For every oracle $\mathcal{O}$, $\mathsf{BPP}^{\mathcal{O}} \subseteq (\Sigma_2^p)^{\mathcal{O}}$.

Time-Bounded Kolmogorov Complexity (original Section 2.1). We fix a time-optimal universal Turing machine U and a time-optimal randomized Turing machine V.

Definition 21. For a string $x \in \{0,1\}^*$, a time bound function $t : \mathbb{N} \to \mathbb{N}$, and an oracle $\mathcal{O}$, we define

$$K^{t,\mathcal{O}}(x) := \min_{d \in \{0,1\}^*} \{|d| : U^{\mathcal{O}}(d) \text{ outputs } x \text{ in at most } t(|x|) \text{ steps}\}.$$

$$rK^{t,\mathcal{O}}(x) := \min_{d \in \{0,1\}^*} \{|d| : V^{\mathcal{O}}(d) \text{ runs in at most } t(|x|) \text{ steps and } \Pr[V^{\mathcal{O}}(d) = x] \geq 2/3\}.$$

We also consider Levin's notion of time-bounded Kolmogorov complexity.

Definition 22. For a string $x \in \{0,1\}^*$ and an oracle $\mathcal{O}$, we define

$$\mathrm{Kt}^{\mathcal{O}}(x) := \min_{d \in \{0,1\}^*, t \in \mathbb{N}} \{|d| + \lceil \log t \rceil : U^{\mathcal{O}}(d) \text{ outputs } x \text{ in at most } t \text{ steps}\}.$$

5.2.2 Barriers for Deterministic Constructions and Lower Bounds for MKtP

In this section, we construct an oracle in EXP relative to which there is no deterministic construction in $2^n/n^{\omega(1)}$ time (even infinitely often), showing that any non-trivial deterministic constructions using PSPACE-relativizing techniques would separate PSPACE from EXP. As an interesting corollary, it is easy to approximate the Kt complexity to an $(1+\varepsilon)$ factor in this oracle world in deterministic $n^{O(\log n)}$ time. Therefore, although it is EXP-hard to approximate the Kt complexity under $(\mathsf{P}/\mathsf{poly})$-truth-table reductions and NP-Turing reductions, any PSPACE-relativizing proof that the Kt complexity requires deterministic $n^{\omega(\log n)}$ time to approximate would separate PSPACE from EXP.

Definition 54. For a constant $\varepsilon > 0$, $\mathsf{Gap}_\varepsilon\mathsf{MKtP}$ is defined to be the promise problem $(\mathcal{YES}_n, \mathcal{NO}_n)_{n \in \mathbb{N}}$, where

$$\mathcal{YES}_n := \{(x,s) \in \{0,1\}^n \times \mathbb{N} : \mathrm{Kt}(x) \leq s\},$$
$$\mathcal{NO}_n := \{(x,s) \in \{0,1\}^n \times \mathbb{N} : \mathrm{Kt}(x) > (1+\varepsilon) \cdot s\}.$$

Our proof relies heavily on the equivalence between non-trivial derandomization and the hardness of $\mathsf{Gap}_\varepsilon\mathsf{MKtP}$ [34]. It is not hard to construct an oracle under which non-trivial derandomization is impossible, which means that a dense subset of the complement of MKtP can be accepted by an efficient algorithm A. Then, we use the worst-case to average-case reduction of [33] to transform A into a worst-case approximation algorithm for $\mathsf{Gap}_\varepsilon\mathsf{MKtP}$.

Theorem 55. There exists an oracle $\mathcal{O} \in \mathsf{EXP}$ such that

1. there is a dense property $Q \in \mathsf{P}^{\mathcal{O}}$ such that every deterministic algorithm that runs in time $2^n/n^{\omega(n)}$ fails to find a string in $Q \cap \{0,1\}^n$ on almost every input length n, and
2. $\mathsf{Gap}_{\varepsilon}\mathsf{MKtP}^{\mathcal{O}} \in \mathsf{DTIME}^{\mathcal{O}}[n^{O(\log n)}]$ for every constant $\varepsilon > 0$.

To show the worst-case to average-case reduction, we use the following pseudorandom generator construction.

Lemma 56 (cf. [33]). For any $d, m \leq 2^n$, $\varepsilon > 0$, there exists a "pseudorandom generator construction"

$$G : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$$

such that for any distinguisher $D : \{0,1\}^m \to \{0,1\}$, if

$$\Pr_{w \sim \{0,1\}^m}[D(w) = 1] - \Pr_{z \sim \{0,1\}^d}[D(G(x,z)) = 1] \geq \frac{1}{2},$$

then

$$K^{\mathrm{poly}(n),D}(x) \leq \exp(\ell^2/d) \cdot m + d + O(\log n),$$

where $\ell = O(\log n)$. Moreover, $G(x,z)$ can be computed in time $\mathrm{poly}(n)$.

Proof of Theorem 55. We construct an oracle $\mathcal{O} \in \mathsf{EXP}$ under which a dense subset of MKtP is in P. We start with $n := 1$, $\mathcal{O} := \emptyset$, and $F := \emptyset$, where F is a set of "frozen" strings. The construction of $\mathcal{O}$ in Stage n is as follows. Let $\mathcal{O}_n$ denote the state of the oracle $\mathcal{O}$ at the beginning of stage n. Let E_n be the set of all strings $x \in \{0,1\}^n$ such that $\mathrm{Kt}^{\mathcal{O}_n}(x) \leq n - c\log n$, where c is a sufficiently large constant (e.g., $c := 3$). Let F_n be the set of all strings $q \in \{0,1\}^*$ such that there exist $k \in [n - c\log n]$ and a description $d \in \{0,1\}^k$ of a Turing machine such that the universal Turing machine $U^{\mathcal{O}_n}$ on input d makes the query q in time $2^{n-k-c\log n}$. Then, we update $F := F \cup F_n$ and $\mathcal{O}_{n+1} := \mathcal{O}_n \cup (\{0,1\}^n \setminus (E_n \cup F))$ and move on to the next stage $n+1$. This completes the description of the oracle $\mathcal{O} := \bigcup_{n \in \mathbb{N}} \mathcal{O}_n$. It is easy to observe that $\mathcal{O} \in \mathsf{EXP}$.

The oracle $\mathcal{O}$ is a dense subset of $\{x : \mathrm{Kt}^{\mathcal{O}}(x) > n - c\log n\}$ in the following sense:

1. $x \notin \mathcal{O}$ for every string $x \in \{0,1\}^n$ such that $\mathrm{Kt}^{\mathcal{O}}(x) \leq n - c\log n$.
2. $x \in \mathcal{O}$ for at least half of the strings $x \in \{0,1\}^n$.

To see the first property, it suffices to show that if $x \in \{0,1\}^n$ satisfies $\mathrm{Kt}^{\mathcal{O}}(x) \leq n - c\log n$, then $\mathrm{Kt}^{\mathcal{O}_n}(x) \leq n - c\log n$ (which implies $x \in E_n$ and thus $x \notin \mathcal{O}$). Assuming that there exists a string d such that $U^{\mathcal{O}}(d)$ outputs x in time $2^{n-c\log n-|d|}$, we claim that $U^{\mathcal{O}_n}(d)$ also outputs x. If not, there exists a query in $\mathcal{O} \setminus \mathcal{O}_n$ made by $U^{\mathcal{O}}(d)$. Let q be the first query in $\mathcal{O} \setminus \mathcal{O}_n$. The same query is also queried during the computation of $U^{\mathcal{O}_n}(d)$. Thus, $q \in F_n$, which implies $q \notin \mathcal{O}$, which is a contradiction.

To see the second property, we bound the size of $E_n \cup F$ at Stage n. The size of E_n is at most $\sum_{k\in[n-c\log n]} 2^k \le 2^n/n^{c-1}$. The number $|F_n|$ of frozen strings in Stage n is at most

$$\sum_{k\in[n-c\log n]} 2^k \cdot 2^{n-k-c\log n} \le 2^n/n^{c-1}.$$

Thus, the number $|F|$ of frozen strings until Stage n is at most $\sum_{n'=1}^{n} |F_{n'}| \le 2^n/n^{c-2}$. Overall, we obtain

$$|\mathcal{O} \cap \{0,1\}^n| \ge 2^n - |E_n| - |F| \ge \tfrac{1}{2} \cdot 2^n.$$

Let $Q := \mathcal{O}$. We prove that $Q \in \mathsf{P}^{\mathcal{O}}$ is a dense property such that every $2^n/n^{\omega(1)}$-time deterministic algorithm A fails to find a string in $Q \cap \{0,1\}^n$ on input 1^n. By the definition of Kt, we have $\mathrm{Kt}^{\mathcal{O}}(A(1^n)) \le O(\log n) + \log(2^n/n^{\omega(1)}) \le n - \omega(\log n)$. Thus, the output $A(1^n)$ of A is not in $\mathcal{O}$.

We now use the worst-case to average-case reduction of [34] to obtain an approximation algorithm $A^{\mathcal{O}}$ for $\mathsf{Gap}_\varepsilon\mathsf{MKtP}$. Let $x \in \{0,1\}^n$ and $s \in \mathbb{N}$ be an instance of $\mathsf{Gap}_\varepsilon\mathsf{MKtP}$. Let $d = k \cdot O(\log^2 n)$, where $k = k(\varepsilon)$ is a sufficiently large constant that will be chosen depending on $\varepsilon > 0$. We may assume without loss of generality that $O(k^2 \cdot \log^2 n) \le s \le n + O(\log n)$ because whether $\mathrm{Kt}(x) \le O(k^2 \cdot \log^2 n)$ or not can be decided in time $n^{O(k^2 \cdot \log n)}$ by an exhaustive search. Let m be a parameter chosen later. Let G be the pseudorandom generator construction of Lemma 56. The algorithm $A^{\mathcal{O}}$ accepts (x, s) if and only if $G(x, z) \notin \mathcal{O}$ for every $z \in \{0,1\}^d$. The running time of $A^{\mathcal{O}}$ is $2^d \operatorname{poly}(n) = n^{O(\log n)}$.

We prove the correctness of $A^{\mathcal{O}}$. Assume that $\mathrm{Kt}^{\mathcal{O}}(x) \le s$. Then, for every $z \in \{0,1\}^d$, we have

$$\mathrm{Kt}^{\mathcal{O}}(G(x,z)) \le \mathrm{Kt}^{\mathcal{O}}(x) + O(d + \log n) \le s + O(d + \log n) \le m - c\log m,$$

where the last inequality holds by choosing a sufficiently large $m = s + O(d + \log n)$. It follows that $G(x, z) \notin \mathcal{O}$ and that $A^{\mathcal{O}}$ accepts (x, s). Conversely, assume that $A^{\mathcal{O}}$ accepts (x, s), which means that $\Pr_{z\sim\{0,1\}^d}[\mathcal{O}(G(x, z)) = 1] = 0$. We claim that $\mathrm{Kt}^{\mathcal{O}}(x) \le (1 + \varepsilon) \cdot s$. Since $\mathcal{O}$ is dense, we have

$$\Pr_{w\sim\{0,1\}^m}[\mathcal{O}(w) = 1] - \Pr_{z\sim\{0,1\}^d}[\mathcal{O}(G(x, z)) = 1] \ge \tfrac{1}{2}.$$

By Lemma 56, we obtain

$$\begin{aligned} K^{\operatorname{poly}(n),\mathcal{O}}(x) &\le \exp(O(\log^2 n)/d) \cdot m + d + O(\log n) \\ &\le \exp(1/k) \cdot m + O(k \cdot \log^2 n). \end{aligned}$$

In particular,

$$\begin{aligned} \mathrm{Kt}^{\mathcal{O}}(x) &\le \exp(1/k) \cdot m + O(k \cdot \log^2 n) \\ &\le (1 + O(1/k)) \cdot m + O(k \cdot \log^2 n) \le (1 + \varepsilon) \cdot m, \end{aligned}$$

where we choose a sufficiently large $k := O(1/\varepsilon)$ and use that $m \ge O(k^2 \cdot \log^2 n)$ in the last inequality. □

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